

Metro Vancouver “Price Sentinel” Network: Food Affordability Mapping Project

About the Project

The Metro Vancouver Price Sentinel Network is an experiential learning initiative under GLOCAL’s Civic Data & Community Empowerment portfolio. This project invites youth volunteers (ages 13 - 30) to collaboratively build a crowdsourced grocery affordability map that helps families, newcomers, students, and low-income residents identify where to shop and how much they can save.

Volunteers contribute across three core tasks - data compilation & research, comparison framework development, and design & visualization. All three tasks launch simultaneously from Week 1, with each feeding into the others across the four-week timeline.

The project is rooted in real Statistics Canada food price data (2017–2025) and addresses a pressing civic issue: British Columbia's food insecurity rate stands at 24.4%, with Metro Vancouver families spending an average of \$17,572 per year on groceries - a 27% increase over five years.

Project Objectives

- Build an interactive, publicly accessible affordability map identifying Metro Vancouver grocery stores offering 15 - 30% savings on essential items.
- Leverage real Statistics Canada price data (2017 - 2025) to document food inflation trends and their disproportionate impact on low-income families, newcomers, students, and seniors.
- Develop volunteers’ technical, analytical, design, and communication skills through hands-on civic data work.
- Build a replication playbook enabling other Canadian cities to launch their own Price Sentinel networks.
- Promote civic literacy around food affordability, data transparency, and the role of government price reporting.

Project Details

Organization: GLOCAL Foundation of Canada

Track: Multi - track (Civic Data & Community Empowerment)

Level of Difficulty: Intermediate

Suggested Hours: 100 - 120 hours over 4 weeks

Age Range: 13 - 30 years

Time Commitment & Task Breakdown

Each volunteer is expected to contribute 25 - 30 hours per week over four weeks (total: 100 - 120 hours). Time allocation varies by focus area and individual role, but a typical breakdown across task types includes:

- **Research & Data Collection:** 25 - 30 hours - Gathering Statistics Canada datasets, web scraping grocery prices, compiling open datasets, maintaining the master database
- **Analysis & Strategy Development:** 20 - 25 hours - Developing the Affordability Index methodology, calculating savings potential, conducting geographic and trend analysis
- **Design & Development:** 25 - 30 hours - UI/UX design, building the interactive map, creating data visualizations and infographics, developing visual brand identity
- **Team Collaboration & Meetings:** 5 - 10 hours - Weekly all - hands meetings, track-specific work sessions, integration syncs, and async coordination via Discord

Note: These estimates reflect the full scope of work across all tracks. Individual volunteers typically focus on 2 - 3 task types aligned with their skills and interests, rather than contributing to all categories.

Core Tasks

Choose the task that fits your skills and interests:

Task 1: Data Compilation & Research

Build the data foundation by finding and organizing grocery price datasets.

What you'll do:

- Download and clean Statistics Canada grocery price data (Vancouver 2017-2025)
- Research and compile pricing information from publicly available online sources
- Build master price database in Google Sheets with 50+ stores and 10-15 items
- Document data sources and create data dictionary

Deliverables:

- Clean CSV file with Vancouver grocery prices
- Master database (Google Sheets or Airtable)
- Data dictionary explaining all columns and sources

Task 2: Comparison Framework Development

Analyze price data to create affordability rankings and savings calculations.

What you'll do:

- Develop Affordability Index methodology (rank stores by price competitiveness)
- Build household savings calculator showing monthly/annual savings potential
- Conduct geographic analysis identifying neighborhoods with affordable options vs. food deserts
- Analyze historical price trends (2017-2025) compared to wage growth

Deliverables:

- 10-15 page Affordability Index report with rankings and analysis

- Savings calculator tool (Excel/Google Sheets)
- Trend analysis charts and geographic maps

Task 3: Design & Visualization

Build the interactive map and create visual materials.

What you'll do:

- Build interactive grocery affordability map using Google My Maps
- Color-code stores by affordability (green = budget-friendly, yellow = average, red = expensive)
- Create 5-7 data visualization infographics for social media
- Design simple brand identity (colors, fonts, logo)
- Test map with 3-5 users and revise based on feedback

Deliverables:

- Live, shareable map link with 20-30 stores plotted
- Embed code for websites
- 5-7 PNG infographics (sized for social media)
- One-page brand style guide

Track 4: Content & Communications

Goal: Turn insights into clear, useful, and accessible content

What you'll do:

- Write a short report summarizing findings and methodology
- Create simple community guides (e.g., student savings, newcomer tips)
- Develop and write social media content highlighting key insights
- Write stories or case studies showing real-world impact

Deliverables:

- Final report (clear and accessible)
- 3-5 short community guides
- Social media content (posts + key messages)
- 2-3 short stories or case studies

Data Sources

All tasks begin with Statistics Canada Table 18-10-0245-01 (Vancouver grocery prices 2017-2025). Additional datasets include:

- BC Census income data by neighborhood
- Consumer Price Index (food inflation rates)
- Labour Force Survey (wage data)
- BC minimum wage history
- CMHC rental market data

All data sources include direct download links in the resources guide.

Learning Outcomes

By completing this project, volunteers will:

- Gain practical skills in data research, analysis, GIS mapping, or data visualization
- Learn to translate complex data into accessible public-facing tools
- Understand connections between food affordability, income inequality, and housing costs in Metro Vancouver
- Develop remote collaboration and project management skills
- Build portfolio pieces demonstrating applied skills for employment or education applications